Relapse Thresholds in Quiescent Cancer Cells Dormancy: A Delay Differential Equation Model for Tumor Growth

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Abstract

Mathematical models are valuable tools for understanding tumor growth dynamics. This paper proposes a novel extension to an existing delay differential equation model that incorporates the immune system's response and a cell cycle-specific drug. The extended model distinguishes three key populations: proliferating tumor cells, dividing tumor cells (mitosis), and immune cells. The delay term accounts for the time spent by tumor cells in the interphase before entering mitosis. We analyze the model's stability using the argument principle and establish a theorem that links stability to the time delay. Furthermore, we demonstrate the potential for periodic solutions arising through Hopf bifurcations, explored both theoretically and numerically. The paper also addresses the rationale behind excluding the quiescent cancer cell population (G_0) from the base model and discusses the potential impact of their reactivation on tumor recurrence. This comprehensive approach provides a deeper understanding of the interplay between tumor growth, immune response, and cell cycle-specific drugs.

CCS Concepts: • **Computing methodologies** → **Modeling and simulation**; **Continuous mathematics**; **Differential equations**; *Discrete-event simulation*; • **Theory of computation** → *Dynamic systems and chaos.*

Keywords: delay differential equations, tumor growth, quiescent cancer cells, stability analysis, basin of attraction, chemotherapy modeling, mathematical biology

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1 Introduction

Cancer's relentless growth poses a constant challenge to human health. In Canada alone, projections estimate nearly 90,000 people aged 35 and above will lose their lives to this disease in 2024 [3]. This grim reality underscores the critical need for advancements in cancer research, particularly in understanding the growth dynamics of solid tumors.

Conventional mathematical models of tumor growth rely on Ordinary Differential Equation (ODE) models, which capture instantaneous rates of change but fail to represent the inherent time delays of biological processes. Cell cycle progression is one such process: a tumor cell does not immediately re-enter mitosis after completing interphase. The time spent in interphase before division, denoted τ, is a biologically meaningful quantity that shapes how tumors respond to treatment. Delay Differential Equation (DDE) models extend the ODE framework by incorporating such explicit time lags [2], providing a more faithful representation of tumor-immune dynamics.

This paper analyzes a DDE model originally proposed by Villasana and Radunskaya [5] and extended with drug interaction parameters by Villasana and Ochoa [4]. The model tracks four coupled populations: interphase tumor cells T_I, mitotic tumor cells T_M, immune cells I, and drug concentration u. We make the following contributions:

1. We reproduce and extend the published analytical stability analysis of the tumor-free steady state, clarifying the non-monotone dependence of stability on the delay τ and its clinical implications for cell-cycle-specific drugs.
2. We numerically estimate the basin of attraction of the tumor-free steady state under chemotherapy, providing a practical tool for identifying initial conditions that lead to tumor eradication without further intervention.
3. We introduce a novel Quiescent Cancer Cells (QCC) extension incorporating a dormant cell subpopulation,

which is estimated to comprise roughly 20% of total tumor mass and largely invisible to standard chemotherapy, and characterize the threshold quiescence parameters beyond which tumor survival becomes inevitable under the simulated regimen.

The central finding is that QCCs substantially reduce treatment effectiveness. An identical chemotherapy regimen that eradicates tumor cells in the base model fails when quiescent cells are present, because dormant cells survive the drug and repopulate the tumor after treatment ends. The extended basin-of-attraction analysis further reveals that values of the quiescence activation rate a_q and death rate d_q above approximately 0.0025 make tumor survival inevitable for the tested regimen. This has direct implications for treatment design: effective long-term tumor control requires therapies that target both proliferating and quiescent subpopulations.

The remainder is organized as follows. Section 2 presents the base DDE model. Section 3 describes the numerical implementation and baseline behavior. Section 4 analyzes stability and the basin of attraction. Section 5 introduces and analyzes the QCC extension. Section 6 concludes.

2 The Delay Differential Equation Model

The model, originally proposed by Villasana and Radunskaya [?] and extended in [?], tracks four coupled populations: $T_I(t)$ (tumor cells in interphase), $T_M(t)$ (tumor cells in mitosis), $I(t)$ (immune cells), and $u(t)$ (drug concentration).

2.1 Assumptions

All mathematical models of biological systems are simplifications. The key assumptions here are [?]: (1) mitosis is instantaneous relative to the overall cell cycle duration; (2) the drug specifically targets and arrests tumor cells during mitosis; (3) the drug is cytotoxic to both tumor and immune cells, reflecting the bone marrow toxicity common to many chemotherapy agents; (4) the quiescent G_0 phase is excluded from the base model, despite quiescent cells being estimated to comprise roughly 20% of total tumor mass and are largely resistant to chemotherapy. This exclusion is addressed in Section 5; (5) drug concentration decays exponentially, corresponding to a single-compartment pharmacokinetic model.

2.2 System of Equations

$$T_I' = 2a_4T_M - (c_1I + d_2)T_I - a_1T_I(t-\tau) \tag{1}$$

$$T_M' = a_1T_I(t-\tau) - (d_3 + a_4 + c_3I)T_M - k_1(1-e^{-k_2u})T_M \tag{2}$$

$$I' = k + \frac{\rho I(T_I+T_M)^n}{\alpha + (T_I+T_M)^n} - (c_2T_I + c_4T_M + d_1 + k_3(1-e^{-k_2u}))I \tag{3}$$

$$u' = -\gamma u \tag{4}$$

The delay τ in Equations 1 and 2 represents the time spent by tumor cells in interphase before entering mitosis. The term $a_1T_I(t-\tau)$ captures cells that entered interphase τ time units ago and are now transitioning to division. This is the key biological realism that ODE models cannot represent.

In Equation 3, the Michaelis-Menten term models immune stimulation by the tumor: as tumor burden increases, immune response grows up to a saturation ceiling α. The drug term $k_3(1-e^{-k_2u})$ captures immune cell toxicity from chemotherapy.

2.3 Biological Interpretation of Terms

$$T_I' = \underbrace{2a_4T_M}_{\substack{\text{cells entering}\\\text{interphase}\\\text{after mitosis}}} - \underbrace{(c_1I + d_2)T_I}_{\substack{\text{combat \&}\\\text{natural deaths}}} - \underbrace{a_1T_I(t-\tau)}_{\substack{\text{cells entering}\\\text{mitosis (delayed)}}}$$

$$T_M' = \underbrace{a_1T_I(t-\tau)}_{\substack{\text{cells entering}\\\text{mitosis}}} - \underbrace{(d_3 + a_4)T_M}_{\substack{\text{natural death}\\\text{\& exit to } T_I}} - \underbrace{c_3T_MI}_{\text{combat deaths}} - \underbrace{k_1(1-e^{-k_2u})T_M}_{\text{drug effect}}$$

2.4 Parameters

Table 1. Parameter values from [?] (non-dimensional). We use $\gamma := \lambda_2 = 0.85$ following the single-drug-term base model [?].

$\tau = 0.92$	$a_1 = 0.98$	$a_4 = 0.8$	$d_2 = 0.11$
$d_3 = 0.4$	$d_1 = 0.29$	$c_1 = 0.9$	$c_3 = 0.9$
$c_2 = 0.085$	$c_4 = 0.085$	$\rho = 0.1$	$\alpha = 0.2$
$k = 0.036$	$n = 3$	$k_1 = 0.47$	$k_2 = 0.57$
$k_3 = 0.49$	$\gamma = 0.85$		

3 Numerical Implementation

3.1 Software and Solver

The model was implemented in Python using the `ddeint` library, a DDE solver that extends `scipy.integrate.odeint` with interpolated history tracking. At each integration step, `ddeint` reconstructs $T_I(t-\tau)$ by cubic-spline interpolation over the previously computed trajectory. Python was chosen for its native support of both `ddeint` and `matplotlib`. The complete implementation is available at https://gist.github.com/jaxendutta/a0aa1541cf9c2ab399b5f49631055566.

3.2 Initial Conditions and History

DDEs require a *history function* specifying the system state for all $t \le 0$. We use the constant history:

$$(T_I, T_M, I, u) = (1.3,\ 1.2,\ 0.9,\ 0) \quad \forall\, t \le 0,$$

representing a small initial tumor burden with baseline immune presence and no drug. A constant history is an accepted simplification since $\tau = 0.92$ is considerably shorter than the simulation duration [4]. Simulations ran over $t \in [0, 3000]$ with 40,000 uniformly spaced time points.

3.3 Baseline Behaviour

Before introducing drug intervention, we verify that the model reproduces the two canonical qualitative behaviours of tumor-immune dynamics.

Figure 1 confirms both regimes. When initial immune population is insufficient relative to tumor burden, proliferation is unchecked and the tumor grows exponentially. When immune cells are sufficient, they suppress the tumor to a low stable equilibrium. The boundary between these regimes is exactly what the basin-of-attraction analysis in Section 4 characterizes numerically.

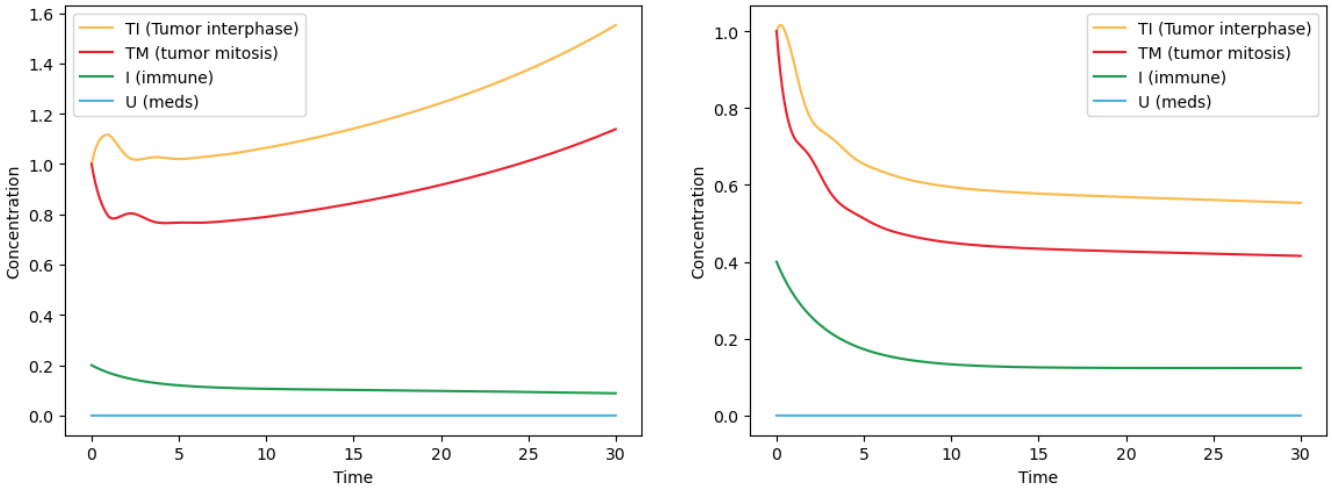

Figure 1. Baseline model behaviour. Left: exponential tumor growth when immune cells are overwhelmed. Right: immune containment, where the tumor stabilizes at a low equilibrium. These two regimes define the basin-of-attraction boundary analyzed in Section 4.

3.4 Chemotherapy Modelling

Drug infusion is modelled as discrete periodic episodes: during active periods, u is held at a target concentration $u_{\max}$; between episodes it decays exponentially at rate γ. The baseline regimen used throughout is $8\times(3$ days active $+\,11$ days rest) with $u_{\max} = 0.7$. This is sufficient to eradicate tumor cells in the base model.

4 Stability Analysis

4.1 Tumor-Free Steady State

The tumor-free steady state $(T_I, T_M, I, u) = (0, 0, k/d_1, 0)$ always exists and represents the target of chemotherapy. Its stability depends on the model parameters and, crucially, on the delay τ [5].

A non-intuitive result from [5] is that stability is not monotone in τ: as τ increases, the tumor-free steady state can switch between stable and unstable a *finite* number of times before becoming permanently unstable beyond a threshold τ^*. This follows from the structure of the characteristic equation associated with the linearized system, which is a transcendental equation in τ with finitely many purely imaginary roots.

Clinical implication. Many cell-cycle-specific chemotherapy drugs work by trapping cells in a particular phase of the cell cycle, effectively increasing τ. If this increase crosses a stability-switching threshold, completing treatment could leave the system in a regime where the tumor-free state is unstable. This means that even if the drug appears to eradicate the tumor during treatment, the system will drift away from the tumor-free state once the drug is withdrawn, causing relapse. This motivates careful attention to drug dosing schedules that maintain τ below τ^*.

4.2 Basin of Attraction: Numerical Estimation

A complementary property of clinical interest is the *basin of attraction* of the tumor-free steady state: the set of initial conditions from which the system converges to tumor-free equilibrium. Once the system enters this basin, for instance, after a course of chemotherapy reduces tumor burden sufficiently, it converges without further intervention. This provides a practical stopping criterion for treatment.

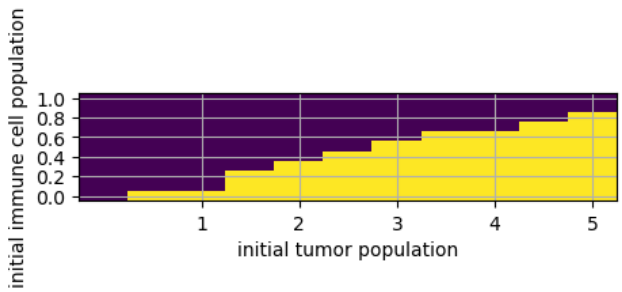

Figure 2. Numerically estimated basin of attraction for the tumor-free steady state under the parameters of Table 1. Horizontal axis: initial total tumor population ($T_I + T_M$, split equally). Vertical axis: initial immune cell population $I(0)$. Purple: converges to tumor-free state. Yellow: tumor survives. The boundary approximates the relapse threshold.

We estimated the basin by testing a grid of initial conditions $(T_I(0) + T_M(0), I(0))$, split evenly between T_I and T_M, and recording whether the total tumor population decreased by end of simulation. Figure 2 shows the result. The purple/yellow boundary approximates the relapse threshold: initial conditions above the boundary converge to tumor-free equilibrium; below it, the tumor survives.

A known limitation of this numerical approach is the possibility of false positives, where a simulation classified as converged due to insufficient duration may eventually show tumor regrowth given a longer time horizon.

5 Extension: Quiescent Cancer Cells

5.1 Motivation

The base model's exclusion of the Quiescent Phase (G_0) phase is a known simplification. QCCs are a dormant subpopulation estimated to comprise roughly 20% of the total tumor cell population [5]. Because the drug in the base model targets cells during mitosis, QCCs are largely invisible to treatment. They survive the chemotherapy course and can reactivate to repopulate the tumor after treatment ends [1]. This mechanism is linked directly to tumor dormancy and late relapse, motivating the extension.

5.2 Extended Model

We introduce a fifth population $Q(t)$ of QCCs, governed by two new parameters: a_q (rate at which interphase cells enter quiescence) and d_q (natural death rate of quiescent cells). The extended equations for T_I and Q are:

$$T_I' = 2a_4 T_M - (c_1 I + d_2 + a_q)T_I - a_1 T_I(t-\tau) + d_q Q \quad (5)$$

$$Q' = a_q T_I - d_q Q \quad (6)$$

Equations 2–4 remain identical to the base model. The term $a_q T_I$ represents interphase cells transitioning to quiescence; $d_q Q$ represents quiescent cells either dying naturally or returning to active cycling. A separate delay $\tau_q = 30$ governs the history of Q, reflecting the longer timescale of quiescent cell dynamics.

5.3 Effect on Treatment Efficacy

Figure 3 demonstrates the core result: the introduction of QCCs makes treatment substantially more difficult. The identical chemotherapy regimen that achieves eradication in the base model fails in the extended model because quiescent cells survive drug exposure and serve as a reservoir for tumor regrowth.

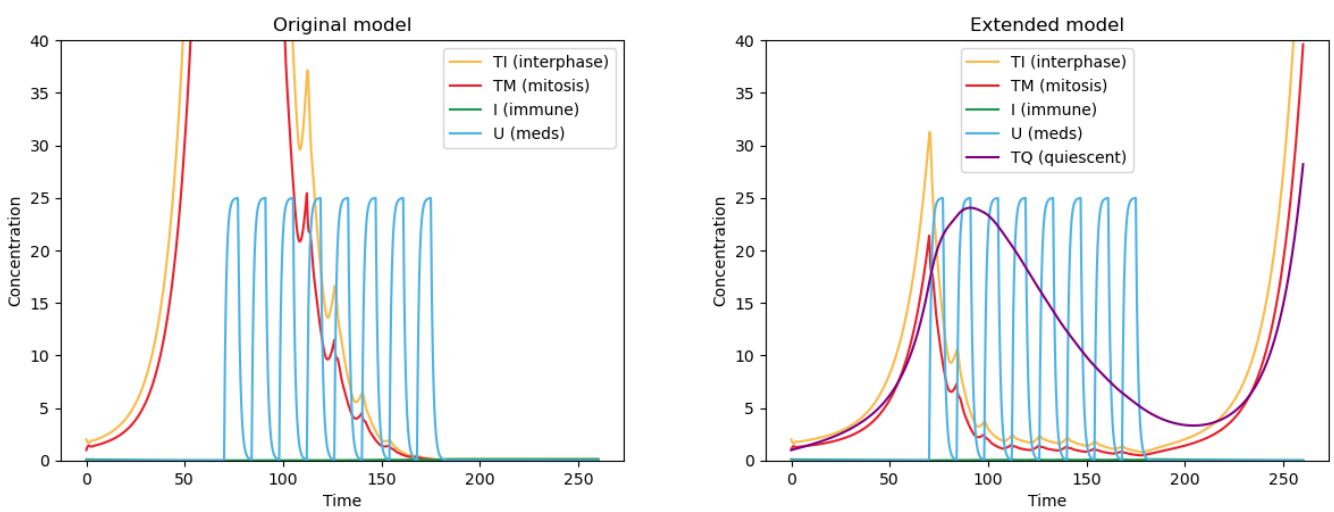

Figure 3. Identical chemotherapy regimen without (left) and with (right) the QCC subpopulation. The regimen eradicates tumor cells in the base model but fails when quiescent cells are present. Dormant cells survive and repopulate the tumor after treatment ends.

5.4 Sensitivity to Quiescence Parameters

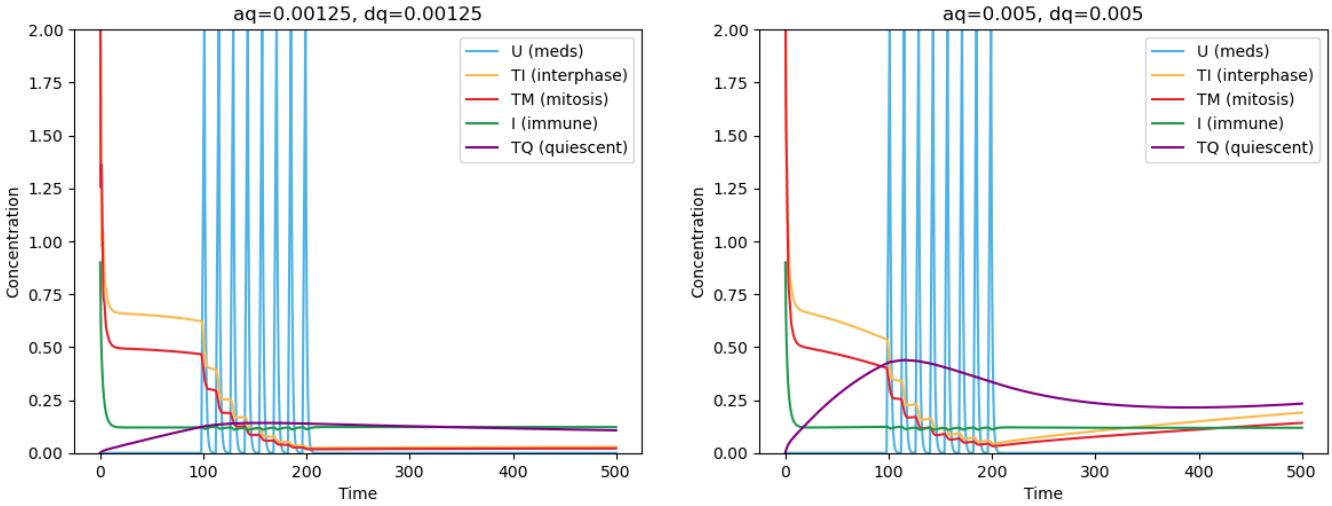

Figure 4. Effect of a_q and d_q on treatment outcome. Left: low quiescence rates allow tumor eradication. Right: higher quiescence rates enable tumor survival after treatment.

The tumor's ability to evade chemotherapy depends critically on a_q and d_q, as shown in Figure 4. To characterize this dependence systematically, we performed an analysis analogous to the basin-of-attraction study of Section 4, varying a_q and d_q over a grid and recording whether the tumor population at the end of the simulation is smaller than immediately after the chemotherapy period.

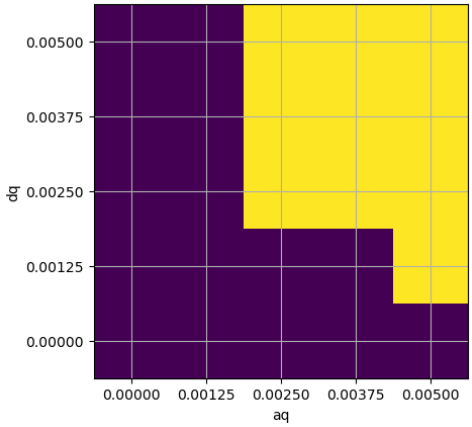

Figure 5. Extended basin of attraction as a function of quiescence parameters a_q and d_q under the baseline regimen ($8 \times$ [3 days + 11 days rest], $u_{\max} = 0.7$, $\tau_q = 30$). Purple: tumor eradicated. Yellow: tumor survives. Values of a_q and d_q above ≈ 0.0025 lead to treatment failure.

Figure 5 reveals a clear threshold at approximately $a_q = d_q \approx 0.0025$. Below this boundary, the drug regimen successfully drives the system into the tumor-free basin. Above it, the quiescent subpopulation is sufficiently active to sustain tumor regrowth despite treatment. This threshold represents a practical relapse boundary: tumors with quiescence dynamics above this level require qualitatively different treatment strategies, specifically agents that can target and eliminate dormant cells in addition to proliferating ones.

6 Conclusion

This paper analyzed a DDE model of tumor growth under cell-cycle-specific chemotherapy, contributed a numerical basin-of-attraction analysis, and extended the model to incorporate QCCs.

The stability analysis of the tumor-free steady state reveals a non-intuitive result: stability does not decrease monotonically with τ. As τ increases, the tumor-free state can switch between stable and unstable a finite number of times before becoming permanently unstable beyond a threshold τ^*. Since cell-cycle-specific drugs effectively increase τ, an improperly dosed regimen could destabilize the tumor-free state and cause relapse after treatment ends, even when the tumor appeared eradicated during treatment. The basin-of-attraction analysis provides a complementary tool: it numerically maps the initial conditions from which the system converges to tumor-free equilibrium, offering a practical criterion for determining when treatment can safely stop.

The QCC extension demonstrates that dormant cancer cells substantially undermine treatment efficacy. An identical chemotherapy regimen eradicates tumor cells in the base model but fails when quiescent cells are present, because dormant cells survive the drug and repopulate the tumor after treatment ends. The extended basin-of-attraction analysis identifies $a_q \approx d_q \approx 0.0025$ as the threshold beyond which tumor survival becomes inevitable under the tested regimen. The model therefore suggests that effective long-term tumor control requires therapies that target both proliferating and QCC subpopulations.

Three directions for future work stand out. First, incorporating spatial tumor heterogeneity would allow the model to represent differences in drug penetration and immune access across tumor regions. Second, patient-specific parameter estimation from clinical data would enable personalized treatment optimization. Third, explicitly modeling the immune system's role in controlling QCC reactivation, which is absent from the current extension, could reveal additional therapeutic levers beyond direct cytotoxicity.

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